

Gate E7 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS with MINOR findings**. Evidence directory: 20_EVIDENCE/E7/.

E7 (attribution / literature) — first pass

Target: 01_manuscript/v1.2_full_rebuild_candidate/PAPER_IV_preprint_v1.2_en.md, sha256 a501d53b527da22f20112db2b05917fb61f46dd8e080faf14b69349b240cc152 (2464 lines). Audit date 2026-09-30. Labels: **SV** = SOURCE_VERIFIED (read in the saved source), **AI** = AUDITOR_INFERENCE, **UV** = UNVERIFIED. Line numbers refer to the manuscript unless prefixed by a source. Page numbers for [15] are those printed in arXiv:2609.20871v1. Quotes from sources are kept under 15 words; everything else is paraphrase.

0. Sources and later-version status

Ref	URL	Version retrieved	Source date	sha256 (saved file)
[15]	https://arxiv.org/pdf/2609.20871v1	v1 (cited)	15 Sep 2026	9654af66...d5fa53 (okechukwu_v1.pdf)
[15] abs	https://arxiv.org/abs/2609.20871	abs page	—	e1ce80ea...b1b0
[22]	https://arxiv.org/pdf/1902.06135v1	v1 (cited)	16 Feb 2019	df77c1d9...6934 (verclos_v1.pdf)
[22] abs	https://arxiv.org/abs/1902.06135	abs page	—	af8ac080...a1b0
[5] PDF	raw.githubusercontent.com/cbde8a0a.../manuscript/main.pdf	commit cbde8a0a	printed 8 Sep 2026	53e15161...3930
[5] README / FORMALIZATION_STATUS	raw.githubusercontent.com/cbde8a0a.../README.md, lean/FORMALIZATION_STATUS.md	commit cbde8a0a	2026-09-08	0e08ab41...0b30b / 05cceb10...93f4
[8]	https://www.erdosproblems.com/81	live, 2026-09-30	page last edited 28 Dec 2025	a02ca0e2...2543
[21]	https://www.erdosproblems.com/forum/thread/81/proof-claims	live, 2026-09-30	claim 201 dated 2026-08-10	69d52613...b329
[23]	Crossref for 10.1137/06064888X	metadata	2008	08dfd5ad...87cb

Full list with URLs and retrieval times: SOURCES_SHA256.txt. Queries: SEARCH_LOG.md.

Later-version status as of 2026-09-30 (SV):

- [15]: the arXiv submission history lists only v1 (15 Sep 2026 16:25:32 UTC), 24 pages, with no journal reference.
- [22]: the history lists only v1 (16 Feb 2019), with no journal reference or DOI. Crossref and web search found no journal version (UV that none exists; the dblp query failed).
- [5]: main HEAD is still cbde8a0a (committed 2026-09-08T10:47:29Z; repo pushed at 2026-09-08T12:53:39Z). Tag v0.1.0-proof-claim points to the same commit. The only other branch is codex/revise-manuscript-lean-audit at 9944194c, an ancestor merged via PR #1. manuscript/main.pdf has the same sha256 at the cited commit, at main and in the release asset digest. **Nothing has changed since the cited commit.**
- [8]: Problem #81 is still marked **OPEN** on the site.

A. Theorem C (1.82–97) vs [15, Theorem 1.1] — component table

Component	[15] v1 (SV, p.1–2, 5–6, 19–20)	Manuscript v1.2	Verdict
Graph class	s-simplicial: $d(v) - \omega(N(v)) \leq s$. $\text{rsd}(G)$ is the least s such that, in every induced H and outside every proper clique of H , there is an s -simplicial vertex. "The empty clique is allowed." (p.2)	1.84: in every induced U , outside every prescribed clique $R \subseteq U$, some vertex's neighbourhood contains a clique omitting $\leq s$ neighbours. The empty root is allowed and the definition is credited to [15, §1].	OK (equivalent formulation, attributed)
Target	$F_n = \lfloor n(n+1)/6 \rfloor$, $Q_s(n) = F_{\lfloor n+s \rfloor} - C(s+1, 2)$ (1.1)	$Q_s(n) = M(n+s) - C(s+1, 2)$, with notation credited to [15, (1.1)] (1.84, 1.90)	OK
Quantifiers	$\forall s \exists K_s \geq 0, N_s$. The bound $cp \leq Q_s + K_s$ holds at all orders; for $n \geq N_s$ the maximum is exactly $Q_s(n)$.	$\forall s \exists N_s \forall n \geq N_s$: $cp \leq c_4 \leq Q_s(n)$. Eventual only.	OK. The eventual scope is stated; the all-orders clause is covered in §8.1 (see G).

Piece size	Statement is for cp only. The explicit constructions (Lemma 2.5, Lemmas 4.1–4.3, (5.11)) use only edges and triangles. Lemma 4.2 is stated for q_3 .	c_4 (pieces of order ≤ 4)	See A.2. The stated strengthening is literally correct.
Attainment	Lemma 2.5, at every sufficiently large order	l.93: attainment for each sufficiently large order, also for unrestricted cp	OK. Credited at l.97.
Extremal classification	Equality holds exactly for $(K_{\{k-s\}} \sqcup K_s) \vee K_{\{n-k\}}$, with k nearest to $(2(n+s)+1)/6$ (l.3). Complement bars confirmed visually from the rendered p.2.	(6.18): $cp = Q_s \Leftrightarrow c_4 = Q_s \Leftrightarrow G \cong (K_{\{k-s\}} \sqcup I_s) \vee I_{\{n-k\}}$, same k (l.1001–1004)	OK. Same family, credited at l.97 and l.1014.
Witness family	book graphs $(K_{\{k-s\}} \sqcup K_s) \vee K_b$	same (Appendix E witness)	OK
Constants	K_s, N_s existential	N_s explicit (tower, Cor. 6.6)	OK, no overclaim (l.1126)

A.1 Role of the cutoff L in [15] (SV)

- **§3 (p.6).** $L \geq 3$ is fixed and q_L and q_L^* are defined. Lemma 3.1 requires $L \geq 4$. Its final assertion is $q_L^* \leq M_n + n \cdot \text{rsd}(G)$.
- **Theorem 3.3 (p.8–9).** For every ε , it produces an integer $L \geq 4$ and constants. In the proof, ξ is chosen first and then L , so that $33(\xi + 1/[2(L-1)]) < \varepsilon/4$. Hence L grows as $\varepsilon \rightarrow 0$.
- **Lemma 3.4 (p.10).** For every fixed L and $\zeta > 0$, $q_L \leq q_L^* + \zeta|G|^2$ for all sufficiently large graphs. The proof bundles templates J and uses Haxell–Rödl/Yuster.
- **Proof of Theorem 1.3 (p.11).** L, δ_0, η_0 are taken from Theorem 3.3 ("The cutoff L is now fixed."). Lemma 3.4 is then applied with error $\eta_0 n^2/2$. Its role is to turn a lower bound on cp (through $q_L \geq cp$) into a lower bound on q_L^* .
- **Proposition 5.4 (p.17) and Theorem 1.1 (p.19).** Prop. 5.4 calls Theorem 1.3 to obtain the cliques needed by Lemma 5.1. Theorem 1.1 uses Prop. 5.4 in a smallest-counterexample argument. L appears nowhere else.
- **(1.5), p.11.** The source says to "use $L = 4$ in Lemmas 3.1 and 3.4". This is the only place where $L = 4$ is chosen.
- **Verdict on l.1255.** The statement is **accurate** (SV): Theorem 3.3 chooses L from the accuracy; Prop. 5.4 and Theorem 1.1 reach it through Theorem 1.3 and Lemma 3.4; and $L = 4$ is used only for (1.5).

A.2 Is the order-four strengthening trivially available from [15]? (AI)

- **What [15] proves literally.** It proves $cp \leq Q_s(n)$ for $n \geq N_s$ and states no bound on c_4 or q_L for non-extremal graphs. In its near-extremal regime (hypotheses of Prop. 5.4: $\delta(G) \geq n/3 - o(n)$ and $cp \geq Q_s$), the construction (5.11) goes through Lemma 4.2, whose conclusion is a q_3 bound. Graphs with $cp < Q_s(n)$ get no constructive treatment: the smallest-counterexample argument only needs the value of cp .
- **Why simple substitution fails.** Running the §5.4 argument with c_4 in place of cp would need Theorem 1.3 under a $c_4 (= q_4)$ lower-bound hypothesis. Lemma 3.4 with $L = 4$ then bounds q_4^* from below. Theorem 3.3, however, needs a lower bound on q_L^* for a large $L = L(\varepsilon)$, and $q_L^* \leq q_4^*$ goes the wrong way. So the literal chain does not give $c_4 \leq Q_s$ by substitution.
- **Limit of this finding.** This is a statement about the literal chain only, not a claim that [15]'s method cannot be adapted. The manuscript makes the same disclaimer at l.1245 and l.1255.
- **Verdict.** Calling Theorem C an order-four strengthening of the *stated* bound (l.95, l.1255) is **OK**.

A.3 Mechanism description of [15] in Table 4 / l.1210 and l.1257 — MINOR (AI, grounded in SV reading)

- **What the manuscript says.** Table 4 gives [15]'s positive-gain pieces as " $K_3, \dots, K_L$ in the far transfer", with " cp , via q_L in that branch". Line 1257 says Lemma 3.4 "pays the budget when $\zeta n^2 \leq M(n) - q_L^*(G)$ " and "is used in the branch with the corresponding margin".
- **What [15] actually does.** The proof of Theorem 1.1 has no far-regime branch in which a margin pays for a transfer loss. Lemma 3.4 enters only inside localization (proof of Thm 1.3), where it converts $cp \geq n^2/6 - o(n^2)$ into a lower bound on q_L^* . The one place it serves as an upper bound is (1.5), with $L = 4$. The explicit partitions [15] builds for Theorem 1.1 use only K_2 and K_3 .
- **Why it matters.** The table therefore gives [15] an architecture it does not have. A reader could infer that [15]'s partitions use large cliques. This does not affect priority claims, and it arguably understates [15].
- **Recommendation.** Rewrite Table 4's [15] row and l.1257 to say that Lemma 3.4 is used for localization and for (1.5). Also say that the near-extremal constructions in [15] are edge/triangle constructions.

B. Checks of cited items of [15]

Manuscript claim	[15] v1 (SV)	Verdict
[15, Cor 1.2] is the chordal case (1.43); optimal cores nearest to $(2n+1)/6$ (1.1263)	Cor 1.2, p.2: $cp \leq F_n + O(1)$; eventual max F_n ; equality exactly $K_k \vee K_{n-k}$, k nearest to $(2n+1)/6$	OK
[15, Thm 1.3] structural sequence result (1.43, 1.1261)	Thm 1.3, p.4: $rsd = o(n)$, $cp \geq n^2/6 - o(n^2)$ give cliques A_j with $\ A_j\ = n/3 + o(n)$ and $D_A + e(G-A) = o(n^2)$; plus (1.5)	OK
[15, Cor 5.5] qualitative edit stability (1.43, 1.1259)	Cor 5.5, p.20: for fixed s and ϵ , there are δ, N such that $cp \geq Q_s - \delta n^2$ gives $\leq \epsilon n^2$ edits to form (1.3). Qualitative.	OK
[15, Lemma 3.1] signed-star argument (1.45, 1.1070, F.2 1.2150)	Lemma 3.1, p.7 (§3.1 "A maximal positive clique-star"). The manuscript's (F.2) rows, the $\alpha = 5c/12$ split and the envelope $c(5n-4c-1)/12$ match [15] p.7–8. The manuscript adds the capped exception charge $\Sigma \min(s, j)$.	OK, adequately attributed ("cited as a method", 1.1070)
[15, Lemma 3.4] $q_L \leq q_L^* + \zeta n^2$ (1.1257); exact fractional partition $\Leftrightarrow$ gain model	Lemma 3.4, p.10, including the "equivalently a fractional packing" remark in its proof	OK for the statement; mechanism framing MINOR (A.3)
[15, Theorem 3.3] $n/6 + 1/24$ term (1.1263)	p.9: $q \leq (\delta_0 + \eta_0)n^2 + n/6 + 1/24$	OK
[15, §7] quantitative version of Cor 5.5 (1.1259)	§7, p.23: asks for the optimal dependence of edits on $Q_s(n) - cp(G)$, n and s . Notes that the proof gives only qualitative stability.	OK
Table 6 (1.1249–1251): signed dual Thm 3.3; Haxell–Rödl + joint templates Lemma 3.4; list colouring + exceptions §§4–5	Thm 3.3 uses the signed dual; Lemma 3.4 uses bundled templates J with Haxell–Rödl/Yuster; §4 uses Galvin's list-edge-colouring theorem and §5 a bounded exceptional set	OK
Order-three obstruction (Prop D.2, 1.95, 1.1748–1761)	Neither [15] nor [5] makes any claim about global order-3 sufficiency or insufficiency. [15]'s near-extremal constructions are order ≤ 3 (Lemma 4.2 is a q_3 bound). [5] builds a K_2/K_3 partition near the extremal family ((5.8): $cp \leq 3 \leq M(n)$) and needs K_4 only through the fractional far functional (p.2 remark on complete graphs).	OK. No contradiction. The manuscript restricts the claim to $s = 0$ and uniformity (1.95).

C. Theorem C' / Cor 6.3 / Prop 6.3a / Prop 6.4 / Thm 6.5 vs [15] — component table

Aspect	[15]	Manuscript	Verdict
Defect regime	Cor 5.5: fixed s . Thm 1.3: $rsd = o(n)$.	C' / Cor 6.3: fixed s . Thm 6.5: $s_j = o(n, j)$.	Parallel structure, correctly matched
Near-extremality hypothesis	on cp	on c_4 ($c_4 \geq Q_s - \delta$). This is implied by the cp hypothesis, so it is a weaker hypothesis and a stronger theorem.	See C.1
Quantitative vs qualitative	Cor 5.5 qualitative (ϵ, δ); §7 explicitly says there is no rate	$d_E \leq A_s \delta$ (non-optimal core) and $d_E \leq A_s \delta + n \vee ((1+4A_s)\delta)$ (optimal size, 6.17); explicit γ_s, N_s stab	Positioning as a "quantitative form" of Cor 5.5 (1.43, which explicitly includes Cor 6.3) is accurate
Square-root lower scale	none	Prop 6.3a family	No conflict with [15]
Partition control	[15] contains no statement about the structure of near-optimal partitions (whole text read)	(6.13), (6.26): one root controls all partitions	"adds simultaneous partition control" (1.43, 1.1261) accurate
Linear / quantitative stability in [15]?	None. The only quantitative objects are Lemma 3.1 / (1.5) and Thm 6.2 (finite signed bound).	—	Manuscript does not misstate [15] here
Thm 1.3 vs Thm 6.5	cliques A_j , $\ A_j\ = n/3 + o(n)$, $D_A + e(G-A) = o(n^2)$	R_j , $\ R_j\ = n/3 + o(n)$, $d_E(G_j, S_{\{R_j\}}) = o(n^2)$, with weaker c_4 hypothesis	"recovers the structural conclusion" accurate
(1.5) vs Prop 6.4 + (6.24)	Stated for cp : $(2n+1)^2/24 + n \cdot rsd + \epsilon(n)n^2$ for every $n \geq 8$, with $\epsilon(n) \rightarrow 0$ graph-independent. Its proof uses $L = 4$ in Lemmas 3.1 and 3.4, so it proves a $q_4 = c_4$ bound.	$c_4 \leq M(n-s) + ns (-C(s+1,2) \text{ on } 4s \leq n) + \epsilon n^2$, with $N(\epsilon)$ independent of G and s . Leading s -term $\approx 2ns/3$ vs ns .	See C.2

C.1 Wording error at 1.1259 — MINOR (AI)

- **The wording.** Line 1259 calls the c_4 near-extremality hypothesis "the stronger near-extremality hypothesis on c_4 ". It then says the result also applies when the hypothesis is stated for cp , since $cp \leq c_4$.
- **Why it is wrong.** The hypothesis $c_4 \geq Q_s - \delta$ is implied by $cp \geq Q_s - \delta$, so it is the weaker hypothesis. The sentence contradicts itself.
- **Fix.** Replace "stronger" by "weaker" or " c_4 -form of the".

C.2 "order-four version of its approximation (1.5)" (1.1261) — MINOR (AI grounded in SV p.11)

- **Why the phrase overstates.** [15] states (1.5) for cp, but its one-line proof already uses $L = 4$, so it literally yields a q_4 bound. The manuscript itself acknowledges this at l.1255.
- **Fix.** Describe the contribution as a refined finite-order and s-dependence form of (1.5), noting that [15]'s proof of (1.5) already works with $L = 4$.

D. [22] de Joannis de Verclos (SV unless noted)

- **What [22] proves.** Theorem 1 (p.2): chordality is testable with query complexity $O(\epsilon^{-37})$, i.e. polynomial ("easily testable"). The contrapositive is a polynomial removal-type statement. The manuscript's wording "polynomial chordal-removal development" (l.1138) is fair (AI).
- **Lemma 3 (p.3).** It assumes $\epsilon > 0$ and $n \geq \epsilon^{-1}$, and concludes that G is $6\epsilon^{1/2}$ -close to chordal. The manuscript (F.10, l.2287) says constant 8 instead of 6, with no $n \geq 1/\epsilon$ condition, and "neither uniformly stronger". This matches. **OK**
- **Lemma 4 (p.3).** It is the Gyárfás–Hubenko–Solymosi large-clique lemma, cited within [22]. Line 2280 says the manuscript "replaces the use of [22, Lemma 4]" and "is not presented as a new version of the full theorem cited there". **OK.** The manuscript could name GHS explicitly; optional.
- **Lemma 11 (p.10–14).** It concerns pinned chordal graphs, with T having at most k leaves and $m_{11} = 2^{40} \epsilon^{-4} k^6 \ln^4 k$. Its proof has three parts that match the manuscript's three cases:
 - Claim 3: every colouring has $\geq \delta n^2$ conflicts, so Theorem 2 (set colouring) applies. This is l11_caseA.
 - Claim 5: a section $G[L_i, R_i]$ far from M_2 -free gives an induced C_4 w.p. $\geq 1/2$. This is l11_caseB1 / claim5_count.
 - Claim 6 and the gluing construction: F chordal and $|E(F) \Delta E(G)| \leq \delta n^2 + (\epsilon/2)n^2 \leq \epsilon n^2$. This is l11_caseB2 / (F.12).

The manuscript's attribution ("retain their roles from the removal strategy of [22]"; no claim that the published Lemma 11 has the modified statement, l.2341) is **accurate**.

- **Table 10 (l.2291–2300).** Four claims were checked against [22]:
 - [22]'s pinned representation is on the fixed tree T (Definition 2, p.9), so "enlarged class" with a path-preserving map is a correct description of a change.
 - Lemma 11's Claim 1 (gate set of size $\leq 3k$ built from leaves) is correctly identified as "not imported".
 - Theorem 2 has $m_2 = 36k \ln(\max|L_u|)\epsilon^{-2}$, matching "published logarithmic bounds".
 - The manuscript's $m_{11} = \lceil 2^{58}(K+1)^{15}/\epsilon^{12} \rceil$ is larger than [22]'s. The manuscript says so (l.2300).

OK

- **Journal version.** None found (UV). The citation to arXiv v1 is correct and current.
- **Overall verdict D: OK.** The description of [22] is accurate.

E. [5] (N0zoM1z0/erdos-81 @ cbde8a0a)

- **Eventual chordal maximum.** SV, [5] p.2 and p.15: Theorem 1.1 (eventual max $M(n)$), Theorem 9.1 ($cp \leq 4 \leq M(n)$ for $n \geq N$) and Corollary 1.2 (all orders). The manuscript's l.43, l.61 and l.1265 are **OK**.
- **FORMALIZATION_STATUS.md.** It is dated 2026-09-08. Its theorems take `Erdos81.ExternalInputs`. `Inputs` with three fields: `VizingInput`, `HaggkvistJanssenInput` and `PackingTransferInput`. The last returns certified rational primal/dual optima and an attained integral optimum. These are stated to be explicit parameters, not axioms. The manuscript's l.1132 is **accurate**, including "does not assert that the mathematical proof of [5] is conditional on conjectures". Additional context: [5] attributes its transfer to Rohatgi–Urschel–Wellens [RUW21, Thm 3.4/3.6] (p.3).
- **Table 5 section locations** (l.1230–1238, l.1217):
 - §2: external inputs / transfer statement

- §3: integral construction around a root (Vizing + Häggkvist–Janssen list colouring)
- §4: strict root regularization
- §5: quantitative local stability
- §6: monotone copy path
- §8: global stability by first entry
- §9: main theorem, $cp \leq 4 \leq M(n)$

All **match** (SV headings).

- **Strict root regularization (I.1267).** [5] Lemma 4.2 (p.7) gives $cp \leq 3(G) \leq p'q' - C(p',2) - m'/9 - A'/2$ for a new clique P' . This equals $B_n(p') - m'/9 - A'/2$ with $q' = n - p'$. **OK** (primes omitted in the manuscript; trivial).
- **Credits to the series (I.1241).** [5]'s introduction (p.1–2) cites Paper II [TG26a] (fractional functional), Paper III [TG26b] (split linear-error bound) and Cipollini. §6 (p.9) says the copy inequality, clone-class lift and complete-split terminal reduction follow the strategy of [TG26a, §§3–5]. **OK**.
- **Changes since cbde8a0a.** None (see §0).
- **Attribution of [5] as "Anonymous" (I.2428) — MINOR / disclosure note** (SV facts, AI assessment).
 - The printed PDF author is "ANONYMOUS", so the manuscript's choice follows the document.
 - Public attribution exists elsewhere, though. The README credits Morluto with the mathematical proof (GPT-assisted) and N0zoM1z0 with the manuscript and Lean. erdosproblems.com proof claim 285 (2026-09-08) is by "Morluto, Yang Luo, Yinzhen Huang, Grace Lee" and links this repository.
 - The manuscript's own tool reference [14] (Jacobian, github.com/morluto/jacobian, I.2414, I.2446) belongs to the same Morluto. [5]'s README and claim 285 say Jacobian played a substantial role in deriving [5].
 - **Suggestions.** (i) Add a pointer to proof claim 285 in [5]'s reference. (ii) Consider a one-line disclosure that the tool [14] is by an author of the [5] claim. This is not an accusation of dependence; the manuscript's ConeAudit (I.1136) concerns formal namespace exclusion only.

F. Cipollini [21]

- **Anchor and date.** SV: anchor #proof-claim-201 exists and is labelled a "partial proof claimed by Ricky Cipollini". It was submitted 2026-08-10 17:06:33.
- **Content.** The summary claims partition into $n^2/6 + o(n^2)$ cliques and names the $o(n^2)$ -to- $O(n)$ gap as the remaining difficulty. The external link is the same Overleaf ID `thjptfhgnmxc` (the site appends #cc1388).
- **Manuscript.** Lines 47 and 2460 are **accurate**. The Overleaf manuscript was not opened (coverage limit).

G. Historical r2 blocker (Theorem C maximum coincides with [15, Thm 1.1]) — status in v1.2

- **Now resolved (SV, manuscript text).** The citation is explicit in five places:
 - abstract I.25: "recovers the unrestricted extremal family of Okechukwu [15, Theorem 1.1]" and "No priority for the extremal value is claimed"
 - §1 I.43, which lists all three clauses of [15, Thm 1.1] and Cor 1.2 with "No priority is claimed"
 - Theorem C I.84 (definition credited), I.95 (strengthening of [15, Thm 1.1]) and I.97 ("maximum and equality family are those of [15, Theorem 1.1]")
 - §8.1 I.1245
 - I.1014

Blocker resolved: OK.

- **Logic of "Theorems C and C' imply all three clauses" (I.1245) (AI).**
 - *All-orders clause.* For $n < N_s$, $cp(G) \leq c_4(G) \leq e(G) \leq C(N_s,2)$, and $Q_s(n) \geq -C(s+1,2)$ (Q_s can be negative at tiny n for $s \geq 1$). So $K_s = C(N_s,2) + C(s+1,2)$ gives $c_4 \leq Q_s(n) + K_s$ at all orders. **Valid.**
 - *Eventual maximum clause.* Given by Theorem C and the witness. **Valid.**
 - *Equality clause.* Given by Cor 6.3 (6.18), which relies on Theorem C' at $\delta = 0$ plus the unrestricted

lower witness, for n above $\max(N_{s^{\text{stab}}}, N_s)$. **Valid as an eventual statement**, which is also how [15] states it.

- **Wording.** Saying "Theorem C' recovers the equality classification" (l.43, l.109) is slightly compressed, since the proof needs Cor 6.3 and the Appendix E witness. **OK.**

- **Residual — MINOR.** The abstract (l.25) and Theorem A (l.61) name only [5] as an alternative source for the all-orders Erdős-81 bound. [15, Cor 1.2] also states it, in cp form, at every order, and [15]'s abstract says it answers the EOZ question. §1 l.43 does credit this. Adding "[15, Corollary 1.2]" next to "[5]" at l.25 and l.61 would remove any residual asymmetry.

H. Overclaim scan

- **Universal $b = 0$.** Not claimed. It is explicitly disclaimed at l.27, l.795, l.1379 and l.2349. **OK**
- **Universal linear mixed packing gap.** Not claimed. It is disclaimed at l.303 and l.2351. **OK**
- **Small or effective threshold.** None claimed. The thresholds are called "explicit but impractical" (l.27, l.795, l.1126, l.2139). **OK**
- **Unqualified superiority.** None found. The comparisons carry disclaimers (l.1215, l.1239, l.1245, l.1255, l.1267).
- **Ambiguity at l.314 — MINOR/UV.** The line says "The near threshold has been reduced from 10^{32} to $4 \cdot 10^{12}$ ". It does not say whose 10^{32} . [5] uses $N = \max\{10^{32}, T(\eta/2)\}$ ([5] (1.3), p.2). The sentence may refer to an earlier version of the manuscript's own threshold, but readers may read it as an improvement over [5]. Recommend naming the baseline.

Additional antecedents noticed (informational, not cited; no severity)

- erdosproblems #81 forum comments: Woett (19 Jun 2026), an explicit EOZ constant $c \geq 1/133$; JamalAgbanwa (20 Jun 2026), a conditional $n^2/6 + O(n)$ bound under a "weak Clique-Drop Hypothesis" (Zenodo 20778270, not opened).
- Bo Ning, arXiv 2608.11536 / 2609.20305 (cp – cc difference): not related to chordal graphs.
- No 2026 work was found, besides [5] and [15], proving stability or c_4 bounds for rooted simplicial defect. This is not a completeness claim.

Summary of verdicts

Item	Verdict
A (Theorem C vs [15, Thm 1.1]; L-cutoff statement l.1255)	OK
A.3 (Table 4 / l.1257 description of [15]'s mechanism)	MINOR (fix recommended)
B (cited items of [15]; order-3 scope)	OK
C (positioning vs Cor 5.5 / Thm 1.3)	OK
C.1 ("stronger hypothesis" wording, l.1259)	MINOR
C.2 ("order-four version of (1.5)", l.1261)	MINOR
D ([22] description and Lemma 11 case mapping)	OK
E ([5] claims, trust boundary, sections, m/9, A/2, credits; unchanged HEAD)	OK
E (Anonymous attribution / Jacobian disclosure)	MINOR
F (Cipollini claim 201)	OK
G (r2 blocker)	Resolved (OK); residual abstract/Thm A asymmetry MINOR
H (overclaims)	OK; l.314 baseline ambiguity MINOR/UV

No BLOCKER candidates and no MAJOR findings in E7.

Coverage limits

- **Databases not searched.** No MathSciNet, zbMATH Open or Google Scholar. dblp failed. There was no full-text search of 2026 journals, and "no later version / no journal version" rests on arXiv, Crossref and a general web search only.
- **Linked sources not opened.** The Overleaf manuscript of [21] and Zenodo 20778270 were not opened.
- **[5] scope.** Only README.md and lean/FORMALIZATION_STATUS.md at cbde8a0a were read. docs/VERIFICATION_STATUS.md, the Lean sources and the certificates were not.
- **Depth of checks.** Proofs in [15], [22] and [5] were not re-verified. The comparisons check statements, numbering and the logical role of lemmas. The analysis in A.2 is an inference about the literal chain, not an

impossibility claim.

- **Manuscript coverage.** Only the manuscript lines listed in the brief plus grep-located mentions of [5], [15], [21], [22] and [23] were read. Other appendices were not audited.

E7 — Lead-auditor verification of the delegated first pass

The delegated first pass (E7_FIRST_PASS.md, same model family) was spot-checked by the lead auditor against the saved literal sources `sources/okechukwu_v1.txt` (sha256 feff7419..., derived from PDF 9654af66...) and `sources/verclos_v1.txt`.

Independently confirmed from the literal text of [15] v1

- Definition (p.1–2): s -simplicial = $d_H(v) - \omega(H[N_H(v)]) \leq s$; $\text{rsd}(G) \leq s$: in every induced H and outside every proper clique (empty allowed) an s -simplicial vertex exists. Equivalent to `RoutedDefectAt` ($\exists$ clique $C \subseteq N_U(v)$ with $|N_U(v)| \leq |C| + s \Leftrightarrow \omega(N) \geq |N| - s$). SOURCE_VERIFIED.
- Theorem 1.1 (p.2): constants K_s, N_s ; $\text{cp}(G) \leq Q_s(n) + K_s$ at every order; for $n \geq N_s$ the maximum is exactly $Q_s(n)$; equality exactly for $(K_{\{k-s\}} \sqcup K_s) \vee K_{\{n-k\}}$, k nearest to $(2(n+s)+1)/6$. Corollary 1.2: chordal case, $\text{cp} \leq F_n + O(1)$ at every order, eventual $\max F_n$ with classification. SOURCE_VERIFIED.
- Abstract of [15] states the chordal all-orders bound "answering a question of Erdős, Ordman and Zalcstein".
- Theorem 1.3 (p.3): sequences, $\text{rsd} = o(n)$, $\text{cp} \geq n^2/6 - o(n^2) \Rightarrow$ cliques A_j of size $n/3 + o(n)$ with $D_A + e(G-A) = o(n^2)$; and (1.5) $\text{cp}(G) \leq (2n+1)^2/24 + n \cdot \text{rsd}(G) + \epsilon(n)n^2$ for every graph of order $n \geq 8$.
- Proof of Thm 1.3 (p.11): L fixed from Theorem 3.3 (L chosen from ϵ , p.9 "choose $\xi > 0$ and then $L \geq 4 \dots$ "); "For (1.5), use $L = 4$ in Lemmas 3.1 and 3.4". Lemma 3.4 (p.10): $q_L \leq q^*_L + \zeta n^2$ (fixed L). SOURCE_VERIFIED.
- Proof of Thm 1.1 (p.19–20): minimal-counterexample argument via Proposition 5.4, whose final count (5.11) is produced by the edge-disjoint-triangle constructions of §4 (Lemma 4.1/4.2); Lemma 3.4 enters only through Theorem 1.3's localization. SOURCE_VERIFIED (reading).
- Corollary 5.5 (p.20): qualitative ϵ - δ edit stability for fixed s under $\text{cp} \geq Q_s(n) - \delta n^2$. §7 (p.23) asks for a quantitative version. SOURCE_VERIFIED.
- Only v1 listed on the arXiv abs page retrieved 2026-09-30 12:28 UTC.

Consequences for the manuscript: the v1.2 text now credits [15, Theorem 1.1] for the class, Q_s , the eventual unrestricted maximum and the equality family (abstract l.25, §1 l.43, Thm C l.84–97, Cor 6.3 l.1014, §8.1 l.1245). **The historical r2 blocker F-01 is resolved in this target** (it remains a historical finding for v1.0.1). The c4 upper bound is a genuine statement-level strengthening; neither the manuscript nor this audit asserts that [15]'s method cannot yield it.

Confirmed MINOR items from the delegated pass (lead auditor agrees)

- E7-M1 Table 4 (l.1210) and l.1257: describing [15]'s use of $K_3 \dots K_L$ as a "far transfer" branch that "pays the budget ... in the branch with the corresponding margin" does not match [15]: Lemma 3.4 is used for localization (Thm 1.3) and for (1.5); Thm 1.1's partitions come from the §4–5 triangle constructions.
- E7-M2 l.1259 "stronger near-extremality hypothesis on c4": logically the weaker hypothesis (see E1-O3).
- E7-M3 l.1261 "order-four version of its approximation (1.5)": [15]'s own proof of (1.5) already uses $L = 4$; only its stated conclusion is for cp . Wording could mislead (OBSERVATION/MINOR).
- E7-M4 Abstract l.25 and Thm A l.61 name only [5] as an alternative source of the all-orders chordal bound; [15, Cor 1.2] (and [15]'s abstract) also claim it. §1 l.43 does credit [15, Cor 1.2], so this is MINOR.
- E7-M5 [5] authorship shown as "Anonymous"; the same group authored the Jacobian tool [14] used by the author. Transparent policy, but a one-line provenance note would help (MINOR; same as r2 E7-C6).
- E7-M6 l.314 "reduced from 10^{32} to $4 \cdot 10^{12}$ ": the prior value's owner is unstated; [5] also uses 10^{32} (per the delegated pass; not re-verified by the lead auditor) — clarity.

[22] spot checks

- Lemma 3 (p.3): $n \geq \epsilon^{-1}$, $G[Y]$ chordal, $p_G(v) \leq \epsilon n^2$ on $X \Rightarrow 6\epsilon^{1/2}$ -close to chordal. Manuscript (F.10) states constant 8 and no $n \geq 1/\epsilon$; comparison accurate. SOURCE_VERIFIED.
- Lemma 11 (p.10): tree T with at most k leaves, G_0 of size $\leq k$, subsets sampled uniformly of size $m_{11} = 2^{40} \epsilon^{-4} k^6 \ln^4 k$. The manuscript's adaptation (sequences with replacement, K nodes, enlarged pinned class, $m_{11} = \lceil 2^{58}(K+1)^{15}/\epsilon^{12} \rceil$) is described as an adaptation, not a literal statement. Accurate.
- Only v1 on arXiv (abs page 2026-09-30).

E7 first-pass status: PASS with MINOR findings (no attribution blocker). Coverage limits: arXiv API, Crossref, GitHub API, erdosproblems.com, one general web search; no MathSciNet/zbMATH/Scholar; Overleaf manuscript of [21] not opened; [5]'s Lean sources not inspected. Not a complete novelty search.

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.